

Ch. 1. Introduction

"beautiful syllabus"

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→ long term issues? (Poverty, Education, Healthcare)
 ↳ solution: growth in economy

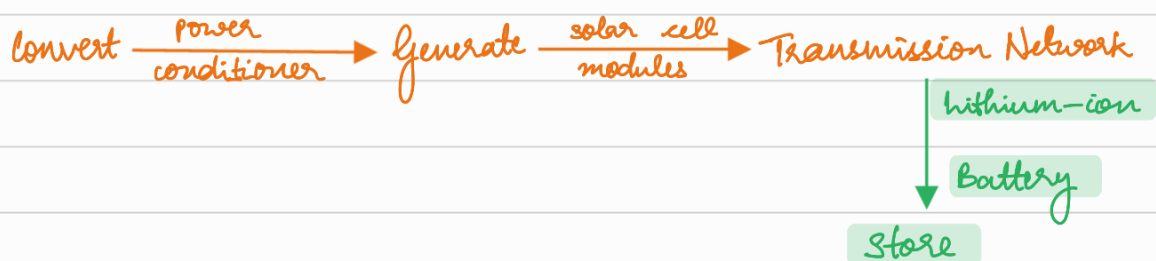
→ Without compromising the ability of future generation ability to meet needs, without causing a lot of environmental damage & wasting limited resources.
 ↳ sustainable development

- conserve, preserve & protect historic, heritage, cultural monuments & locations
- prevent waste of food products
- recover, recycle & reuse materials
- avoid & minimize loss, breakage or wastage
- save energy
- develop alternate sources of energy
- store for the future - generations & needs
- Encourage eco-friendly products
- allows resources to replenish & regenerate by natural means
- Ban roads and temples into deep forest locations
- prefer mass transportation systems to individual vehicles
- Supplements, compliments & support social welfare programmes
- adopt simple living
- Efficiently manage system

→ Father of green chemistry ⇒ Paul Anastas

- champion of sustainability science & innovation for environmental protection.
- worked on designing & manufacturing of chemicals that are non-hazardous and environmentally benign.

Next-generation solar power system



- Green Chemistry $\Rightarrow$
- ① It is safe
 - ② It conserves raw materials & energy
 - ③ It is more cost effective than conventional methods

→ There are 3 main ways to make chemical processes "greener" :-

- ① Redesign production methods to use different, less hazardous starting materials.
- ② Use milder reaction conditions, better catalysts & less hazardous solvents.
- ③ Use production methods with fewer steps & higher atom economy.

Definition:- Sustainable process is utilization of a set of principles that reduces or eliminates the use of generation of hazardous substance in the design, manufacture & application of chemical products.

→ 12 Principles of Green Chemistry

- ① Prevention (treat/clean up waste)
- ② Atom Economy (Synthetic methods to maximize incorporating all materials)
- ③ less Hazardous Chemical Synthesis (design of less/no toxic methods of substances to human environ.)
- ④ Designing safer chemicals (designed to preserve efficacy, reducing toxicity)
- ⑤ Safer Solvents & Auxiliaries (eg:- water, universal solvent)
- ⑥ Design for Energy Efficiency
- ⑦ Use of Renewable Feedstocks
- ⑧ Reduce Derivatives
- ⑨ Catalysis (Catalytic reagents superior to stoichiometric reagents)
- ⑩ Design for Degradation
- ⑪ Real-Time Analysis for Pollution Prevention

15/1/26

① Oxidation - Reduction reactions

② REACTION:- a) Addition

b) Condensation

c) Elimination

d) Rearrangement (Intramolecular or Interamolecular)

* Claissen Rearrangement



16/1/26

→ Benefits of green chemistry

- the environment: products will biodegrade & won't persist in the environment
- human health: products won't cause toxicity to humans
- the economy: novel products which boost competitiveness
- for sustainability: products made from renewable resources
- for science: fundamental new insights



$$\downarrow \text{Risk} = \downarrow \text{Hazardous} \times \text{Exposure}$$

→ Waste Prevention

→ Industry wise waste products produced

Oil refining > bulk chemicals > fine chemicals > Pharmaceuticals
 E-factor grows (left → right)

Industry sector	Annual produce (t)	E-factor	Waste produced (t)
Oil refining	$10^6 - 10^8$	ca. 0.1	$10^5 - 10^7$
Bulk chemicals	$10^4 - 10^6$	< 1 - 5	$10^4 - 5 \times 10^6$
Fine chemicals	$10^2 - 10^4$	5 - 50	$5 \times 10^2 - 5 \times 10^5$
Pharmaceuticals	$10 - 10^3$	25 - 100	$2.5 \times 10^2 - 10^5$

→ Atom Economy

Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.

Q) How does an engineer evaluate a chemical product or process for its effect on human health & the environment?

- #:
- ① starting materials/feedstocks
 - ② Reaction types
 - ③ Reagents
 - ④ Solvents & reaction conditions
 - ⑤ chemical products/target molecules

→ Toxicity to humans :-

- The toxicity of substances to humans has been the focus of extensive testing and has resulted in a no. of large databases.
- The databases contain voluminous information on the levels at which substances will produce a toxic effect in humans.
- Factor, potency is a marker of preferability of one substance versus another.

→ Toxicity to Wildlife :-

- Wildlife covers an enormous no. of species and any evaluation could not hope to include the full range.
- Problematic situation is where a substance may actually be beneficial to one species and detrimental to another.

Example :- Phosphates issues in pool.

→ Evaluating reaction types :-

- Identify proper pathways.
- Thorough literature search of the massive collection of chemical books, journals & other publications will cover proper reports of a given synthetic process.
- The information available to the researcher, in combination with his/her own chemical skills.

→ The various reactive processes available to synthetic chemist can be categorized in a variety of different ways.

Example:- Functional groups are being constructed.

No. of atoms being transformed during the specific synthetic transformation.

→ Transformation may be viewed as addition, substitution, elimination, pericyclic & oxidation/reduction reactions.

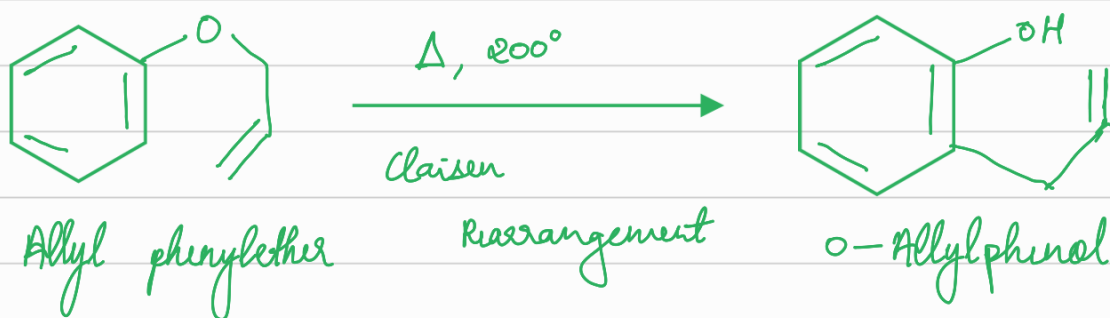
→ Rearrangements :-

→ As the name indicates, are classes of reactions where the atoms comprising a molecule change their orientation relative to one another, their connectivity, their bonding pattern, etc. to yield a new molecule.

→ The primary characteristic of this reaction class from a green chemistry analysis perspective is that the feedstock or starting material and the end-product or target molecule both contain the same atoms.

→ Therefore, there is no intrinsic waste generated necessarily from a rearrangement reaction.

→ From an inherent analysis, a rearrangement reaction is fully atom economical & fully efficient



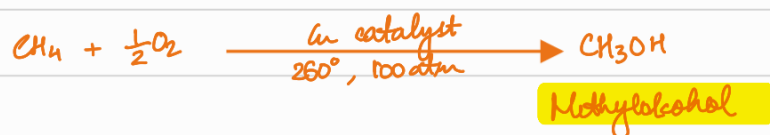
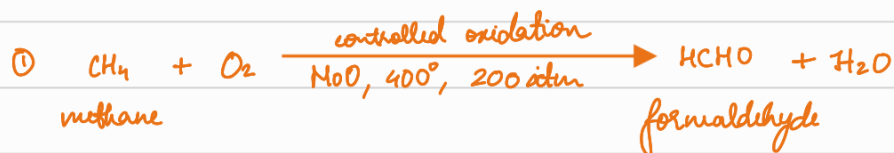
→ Addition reactions:-

- In this general synthetic scheme, multiply bonded, unsaturated substrate is treated with a reactive reagent.
- The addition of bromine to an olefin, ethylmagnesium bromide to carbonyl, hydrogen cyanide to an α, β -unsaturated carbonyl compound serve as example of addition reactions.
- The removal, containment, recycling or disposal of any planned by-products is not necessary, because no such by-products are produced.
- Of course, unplanned by-products that are formed during the process through



Organic Synthesis Using Green Reagents

- Oxygen :- ground-state oxygen



- Singlet Oxygen :-

→ excited state of oxygen

→ has two odd electrons possessing anti-parallel spins



→ can be generated by the oxidation of hydrogen peroxide with alkaline solution of bromine or sodium hydrogen peroxide



- Ozone :-

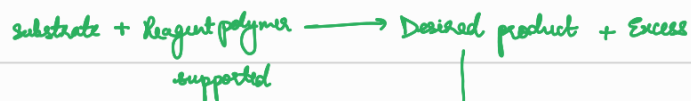
→ blue gas / dark blue liquid used in a mixture w. oxygen

→ generally prepared by passing O₂ thru a silent electric discharge

• Hydrogen Peroxide :-

→ commercially available

• Polymer Supported Reagents



filtration

filtrate contains
desired product

precipitated
polymer supported
reagent (can be renewable)

19/1/26

→ Bluorous Biphasic system

→ Organic synthesis in water :-

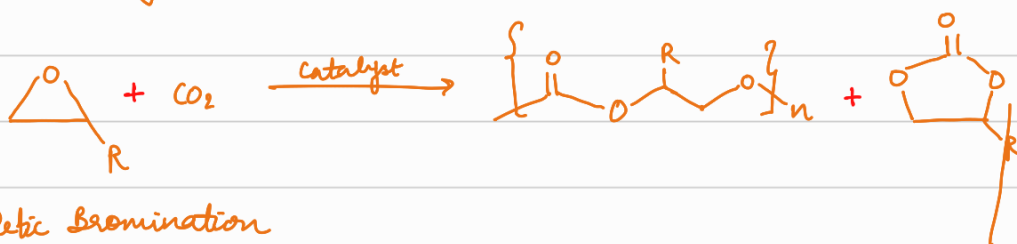
- ① comparatively cheaper solvent.
- ② safe solvent, free of instantaneous reactions.
- ③ soluble, substances can directly dissolve easily.

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* repeated Q's ⇒ supercritical CO₂ works in forming micelle & removing impurities



→ Supercritical Polymerization



→ Synthetic Bromination

→ Ionic liquids [usually are below 100°C]

- Ionic liquids or molten salts are in general defined as liquid electrolytes composed entirely of ions.
- Novel replacement for volatile organic compounds.
- consists of only 2 components (cation & anion), can vary.
- RTIL: room-temp. ionic liquids

23/1/26

→ Kukemengel Condensation

→ Wittig reaction

→ PTC
Nucleophilicity
is enough

Asymmetric dihydroxylation of olefins

1) NaCl dissolves in water

2) Alkyl bromide dissolves in organic solvent

3) PTC grabs Cl^- from water

4) PTC carries Cl^- into the organic layer

5) Cl^- reacts with alkyl bromide

6) PTC returns back to water & repeats the job

→ hence called a catalyst

27/1/26

→ Evaluation of methods to design safer chemicals

→ Atom Economy

→ Percentage yield

$$\% \text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

→ Atom Economy

$$\% \text{ atom economy} = \frac{\text{Mass of desired product(s)}}{\text{total mass of reactants}} \times 100$$

Q) atom economy of H?



A:

$$\begin{array}{l} \text{CO}_2 \longrightarrow 12 + (2 \times 16) = 44 \\ 2\text{H}_2 \longrightarrow 2 \times 2 = 4 \end{array} \quad \left. \vphantom{\begin{array}{l} \text{CO}_2 \\ 2\text{H}_2 \end{array}} \right\} 48 \text{ g}$$

$$= \frac{4}{48} = 0.083 = \underline{\underline{8.3\%}}$$



A: $(12 \times 6) + (1 \times 12) \longrightarrow (12 \times 6) + (1 \times 12) \Rightarrow 84 \longrightarrow 84 \Rightarrow \frac{84}{84} \times 100$
 $\Rightarrow \frac{1}{1} \times 100 \Rightarrow \underline{\underline{100\%}}$



A: $N_2H_4 = (2 \times 14) + (4 \times 1) = 32g$

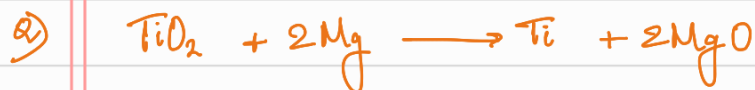
$NaCl = 23 + 35.5 = 58.5g$

$H_2O = (2 \times 1) + 16 = 18g$

tot. mass of reactant $= 32 + 58.5 + 18 = 108.5g$

$\therefore \% a.e = \frac{32}{108.5} \times 100 = \underline{\underline{29.5\%}}$

20/1/26



$Ti = 48g$

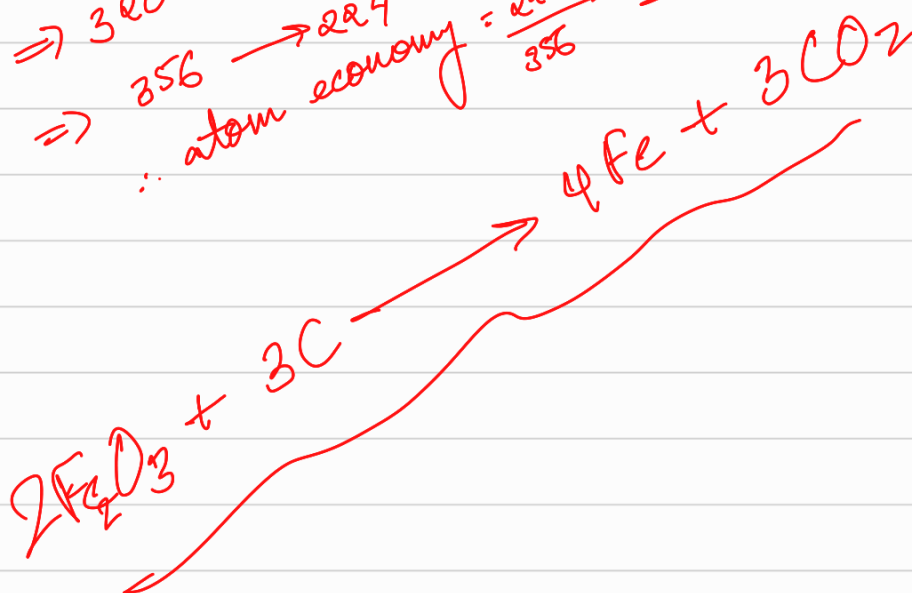
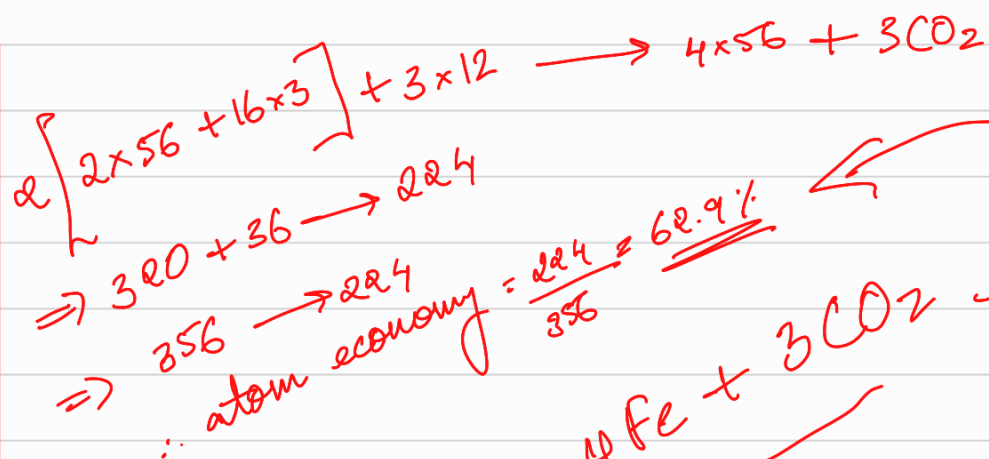
$MgO = 24 + 16 = 40g$

$2MgO = 2 \times 40 = 80g$

tot. $= 48 + 80 = 128g$ (prod.)

$(12 \times 10 + 1 \times 22) = 120 + 22$
 $= 142$

$C_2H_4 = (12 \times 2) + 1 \times 4$
 $= 24 + 4 = 28$
 $\Rightarrow \frac{28}{142} = \underline{\underline{19.7\%}}$



→ E-factor

$$E = \frac{\text{tot. mass of waste}}{\text{mass of final product}}$$

→ Reaction mass

efficiency (RME)

$$\text{RME}\% = \frac{\text{mass of prod C}}{\text{mass of A} + \text{mass of B}} \times 100$$

→ a-efficiency (a.E)%

$$\text{aE}\% = \frac{\text{m.w. of prod.}}{\sum \text{m.w. of reactants}} \times 100$$

→ carbon efficiency

$$\text{CE}\% = \frac{\text{carbon in prod}}{\text{tot. carbon in react.}} \times 100$$

→ Mass Intensity (MI)

$$\text{MI} = \frac{\text{tot. mass in process}}{\text{mass of product}}$$

→ Mass Productivity (MP)

$$\text{MP} = \frac{\text{mass of prod}}{\text{tot. mass of process}}$$



A:-

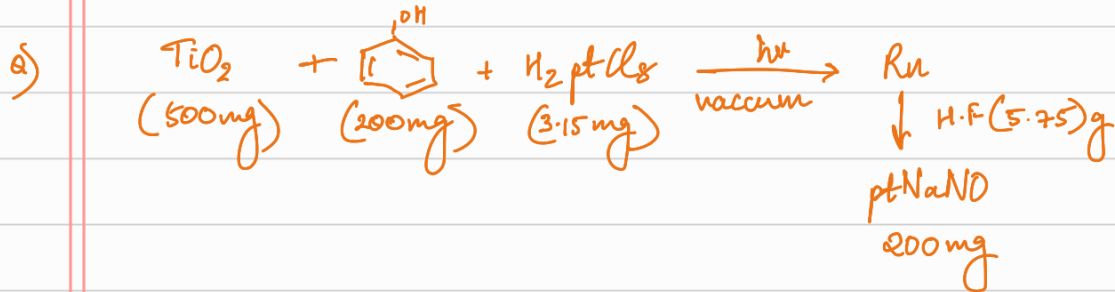
$$40 + 16 = 56$$

$$40 + 12 + (16 \times 3) = 100$$

$$12 + (16 \times 2) = 44$$

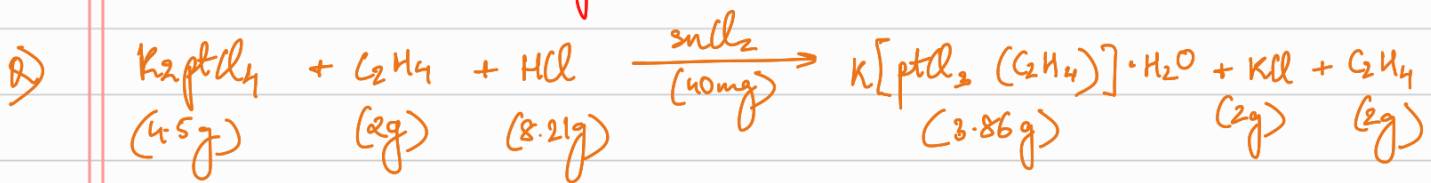
$$100 - 56 = 44$$

$$\therefore E = \frac{44}{56} = \underline{\underline{0.79}}$$



A:- $(500 + 200 + 3.15) = 703.15 \text{ mg}$

final answer $\underline{\underline{= 31.25}}$ $\downarrow$ E-factor



find e-factor.

$$4.5 + 2 + 8.21 =$$

Q) $10.81 + 21.9 = \underline{\underline{32.71 \text{ g}}}$ (React)

Toluene = 500g ; Triethylamine = 15g ; Sulfonate ester = 23.6g

A:-

$$(32.71 + 515) = 547.71 - 23.6 = \underline{\underline{524.11}}$$

$$23.6 \text{ g} ; \therefore \text{E factor} = \underline{\underline{22.21 \text{ kg}}}$$

9/2/26

→

Examples of green reactions

→

Waste Minimization Techniques (Strategy)

(i)

life cycle assessment

(ii)

Environmental Impact assessment

→

purpose of LCA

① compare several opt. for prod. making

② modify a process

③ change packaging

6/2/26

13/2/26



Surface Area of catalysts

$$S.A = \frac{\sigma \cdot m \cdot V_m \cdot N}{(22,400 \times 10^{20}) (m^2 g^{-1})}$$

Q) When a solid of cube form & width side of $l=1\text{ cm}$ is divided into smaller cube of side 0.5 cm .

The original cube has a surface area of 6 cm^2 while the 8 smaller cubes have a total area of?

A:

$$S.A \text{ of original cube} = 6 \times (1)^2 = 6\text{ cm}^2$$

$$\text{no. of small cubes} = \left(\frac{1}{0.5}\right)^3 = 2^3 = \underline{\underline{8}}$$

$$\begin{aligned} S.A \text{ of one small cube} &= 6 (0.5)^2 \\ &= 6 (0.25) \\ &= \underline{\underline{1.5\text{ cm}^2}} \end{aligned}$$

$$\therefore \text{total S.A of all small cubes} = 8 \times 1.5 = \underline{\underline{12\text{ cm}^2}}$$

TOF

includes time factor

$$\text{TOF} = \text{TON} / \text{time}$$

TON

does not include time factor

$$\text{moles of prod.} / \text{moles of catalyst}$$

11/2/26

Phase Transfer Catalyst

- ionic substances usually quaternary ammonium salts, where the size of the hydrocarbon group in the cation is large enough to confer good solubility of the salt in organic solvents.
- meaning cations are highly lipophilic.

**

→ three imp. roles of PTC

- ① Influence of solvent
- ② Influence of cation
- ③ salt effects

20/2/26

Zeolites

Uses:-

- ① Energy Saving Process
- ② with Industrial Applications

Alkylation

Partial oxidation

Catalyst Regeneration

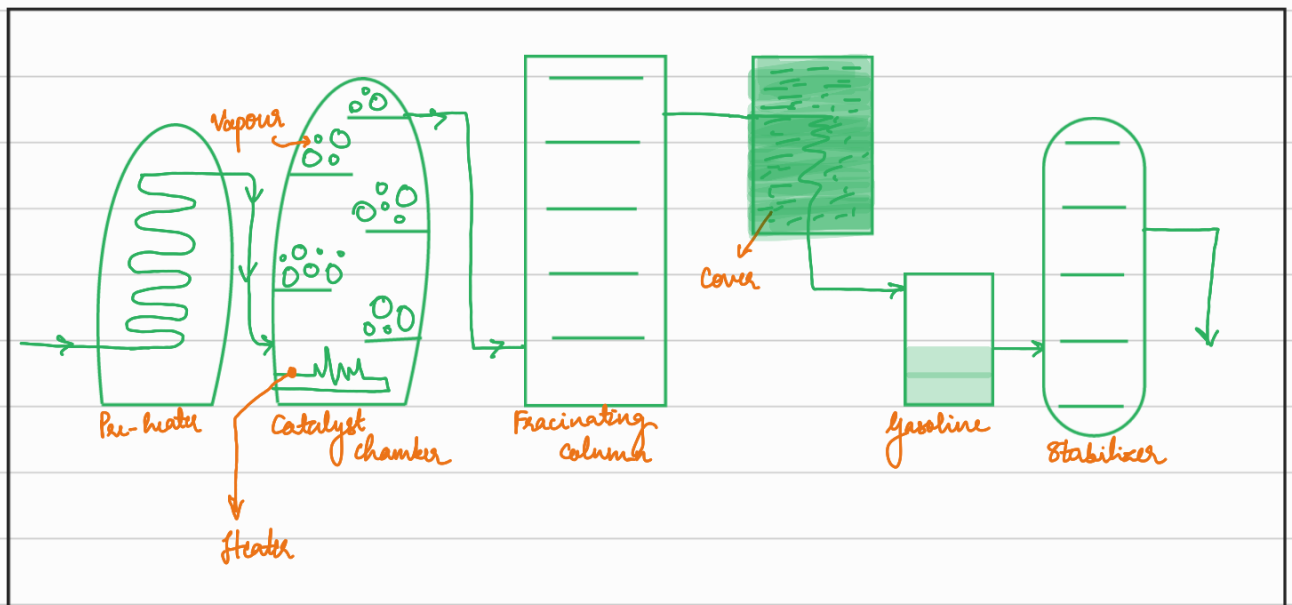
→ sintering

Nickel catalyst

Nickelous
portion till

IA2 next
Friday

Catalytic Cracking



12/2/26

till fixed Bed Catalytic backing → Midterm Portion

→ Fischer - Tropsh Process

→ Hydrogenization of Catalyst

27/1/26
→

Covalent binding of catalyst :-

- The T-CAT unit can often be attached directly to the preformed polymer.
- generally stable link and more than one route to form catalysts. Requirement for covalent attachment can add complexity and

ANY two methods of heterogenization of catalysts / type of attachment

→ Ionic Attachment / heterogenisation :-

- refers to the incorporation of an ionic species into a solid catalyst, resulting in a heterogeneous catalytic system.
- enhances solid to the surfaces to enhance catalytic activity & selectivity.
- Procedure :-
 - ① Impregnation (Immersing the solid support)
 - ② Adsorption (adsorb resulting in ion-exchange or)
 - ③ Drying

→ Supported liquid phase :-

- Relatively simple to carry out w. catalyst soluble in SLP. Otherwise modification of catalyst required to make it soluble in SLP but insoluble in bulk liq. medium.
- key concept is the immobilisation of thin film of liquid on the surface & within the pores of a solid support.

→ Ship in a bottle :-

- Effective immobilisation but restricted range and diffusion difficult

→ Entanglement :-

- delay in the process
- used in nanoparticle catalysis
- simple to do, & polysaccharides generally inexpensive.
- Need for interaction b/w precursor &

2/8/26

→ Properties of polymer support :-

~~XX~~

→ Ziegler-Natta Polymerization :-

13/3/26



Electrocatalysis



HER/ORR vs. OER



Biocatalysis

16/3/26



cleaner Production



IA3: Assignment written in A4 sheet paper

19/3/26



Energy Efficiency

→ air compressor waste heating recycling

→ factory management

→ time control of air dryer



Mining Resource → Prod. → using stage →



LCA & framework



HTK (High Throughput Experimentation)

20/5/26

Q

What is High Throughput Experimentation (HTE)?

The key principle of high throughput experimentation is parallelization. This means that rather than carrying out single experiments one after another, you can run several tests simultaneously.

Steps can be asked for links

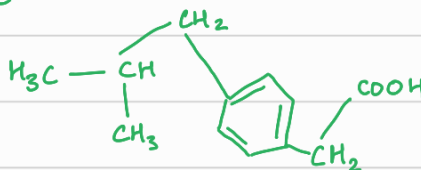
Case Study:- ibuprofen

~ Aspro/Panadol (aka)

~ poor atom efficiency

~ initial synthesis observed the green principles as a disadv.

~



① 2-methylpropyl benzene

② H_2 / Raney Ni

③ Co/Pd [Cotatt / Paladium]

④ NH_2OH

⑤ P_2O_4

⑥ H_3O^+

~ In 6-step method $\Rightarrow$ yield is 90%, prod. is 40%.

but resulted in more by-prod. waste.

~ leading to chem. waste & Energy lost by low efficiency.

~ In 3-step method $\Rightarrow$ yield is 77% where Raney Ni catalyst can be recycled & reused.

→

Cleaner synthesis

It is making a product by novel route that is both commercially advantageous & less of a burden on the environment.

Waves (Micro, ... etc) } not class
Industrial lab's }

23/3/26

Microwave Assisted Organic Systems

- lengths b/w 1 cm & 1 m
- freq. (G): 30 GHz to 300 GHz
- similar to radar & telecommunications
- household/indus. :- 2.45 GHz
- Microwave generator called a magnetron

26/3/26

MW Heating

lexer (reads first char only)
int cost = 10, i.e. = 5 tokens
Relational op. (LE, <=)

Function of Ultrasound

S → A a ? left recursion
A → S c

Operator precedence is not defined
→ multiple parse trees

$E \rightarrow E + T / T$
 $E \rightarrow TE'$
 $E' \rightarrow +TE' | \epsilon$
 $A \rightarrow AA' / B$
 $A \rightarrow BA' ; A' \rightarrow AA' | \epsilon$

left recursion
infinite recursion
in parsing

Ambiguity Multiple parse trees

$S \rightarrow A a / b$
 $A \rightarrow S c / d$

$E \rightarrow E - E / id$
 $E \rightarrow E - E$
 $E \rightarrow id$
 $\rightarrow E$

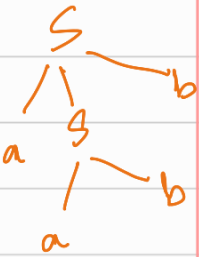
$id \neq id + id$
 $E \rightarrow E - E$
 $\rightarrow id - id$

if (a == b) a = b + 1;

$S \rightarrow \frac{iEtS}{\alpha_1} \mid \frac{iEtSeS}{\alpha_2} \mid \frac{a}{\beta}$

if ([Unmatched parentheses error ..

↓ syntax
↓ by any
Sema
↓



27/3/26

Dimerisation of Maleic Acid to Fumaric Acid

→

Photo Induced Organic Synthesis

30/3/26

→

UV radiations for clean synthesis

→ styrofoam

→ VOC's

→ Electromagnetic Spectrum

2/4/26

→

Cellulose

→

chitin

→

Pharmaceuticals

10/4/26

→

Biopesticides (Introduction)

→

Q1

→ no formula nor weights are given